LibSpace

A Smart Digital Library Management System

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Introduction

- LibSpace is a digital library management system.
- t helps to manage books and users efficiently.
- It replaces manual record keeping.
- It makes library operations simple and fast.

Need for LibSpace:

- Manual system consumes more time.
- High chances of errors in record keeping.
- Difficult to track issued and returned books.
- Need for secure and organized data storage.

Objectives:

- To automate library activities.
- To maintain accurate book records.
- To provide quick search facility.

Key Features:

- Book search option.
- Issue and return management.
- User registration and login.
- Fine calculation system.
- Report generation.

System Modules:

- Admin Module
- Librarian Module
- Student/User Module
- Database Module

Software Requirements:

Node.js and npm installed
Database setup (MongoDB / MySQL)
Code editor (e.g., VS Code)
Web browser (Chrome/Firefox)

Advantages:

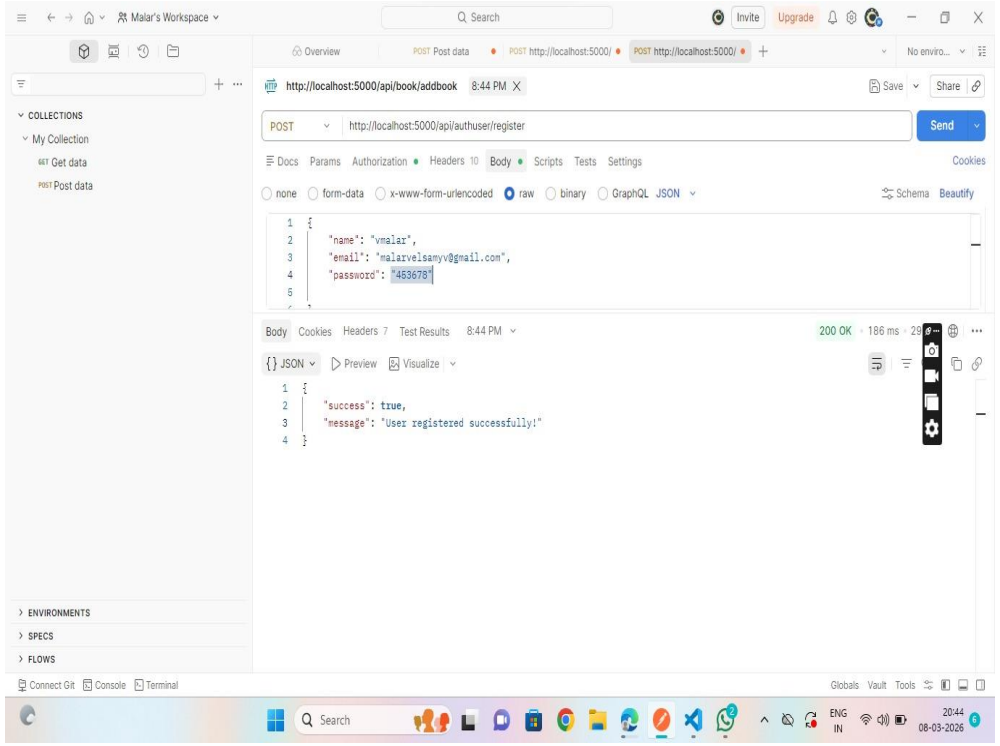
- Saves time and effort.
- Reduces paperwork.
- Secure data storage.
- Easy access to information.
- Improves efficiency.

Test API EndPoints:

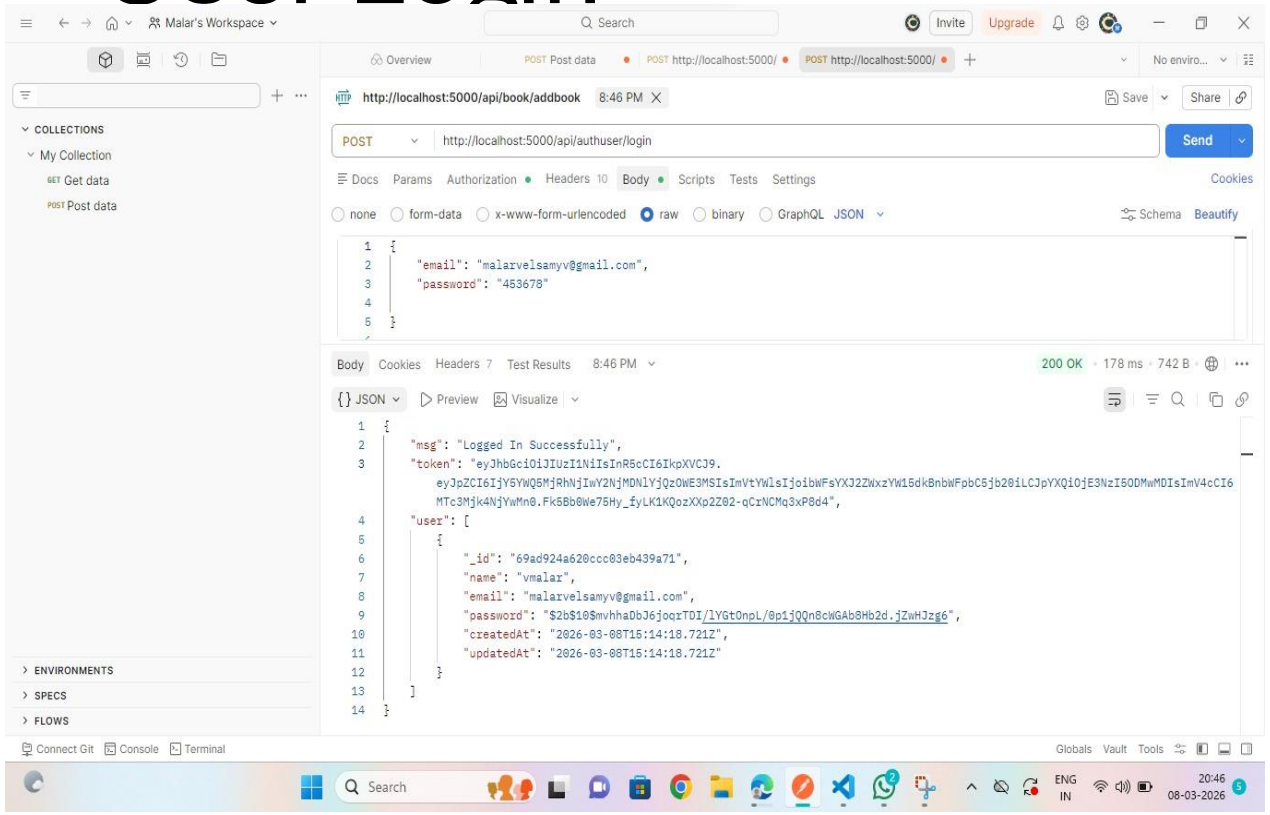
- Use Postman or any API testing tool to test the endpoints
- **User Routes:**
- POST /api/auth/register
- POST → /api/auth/login
- DELETE/api/auth/delete
- **Book Routes:**
- POST/api/book/add
- GET/api/book/
- PUT → /api/book/:id
- DELETE → /api/book/:id
- Base URL:
- http://localhost:5000

Output:

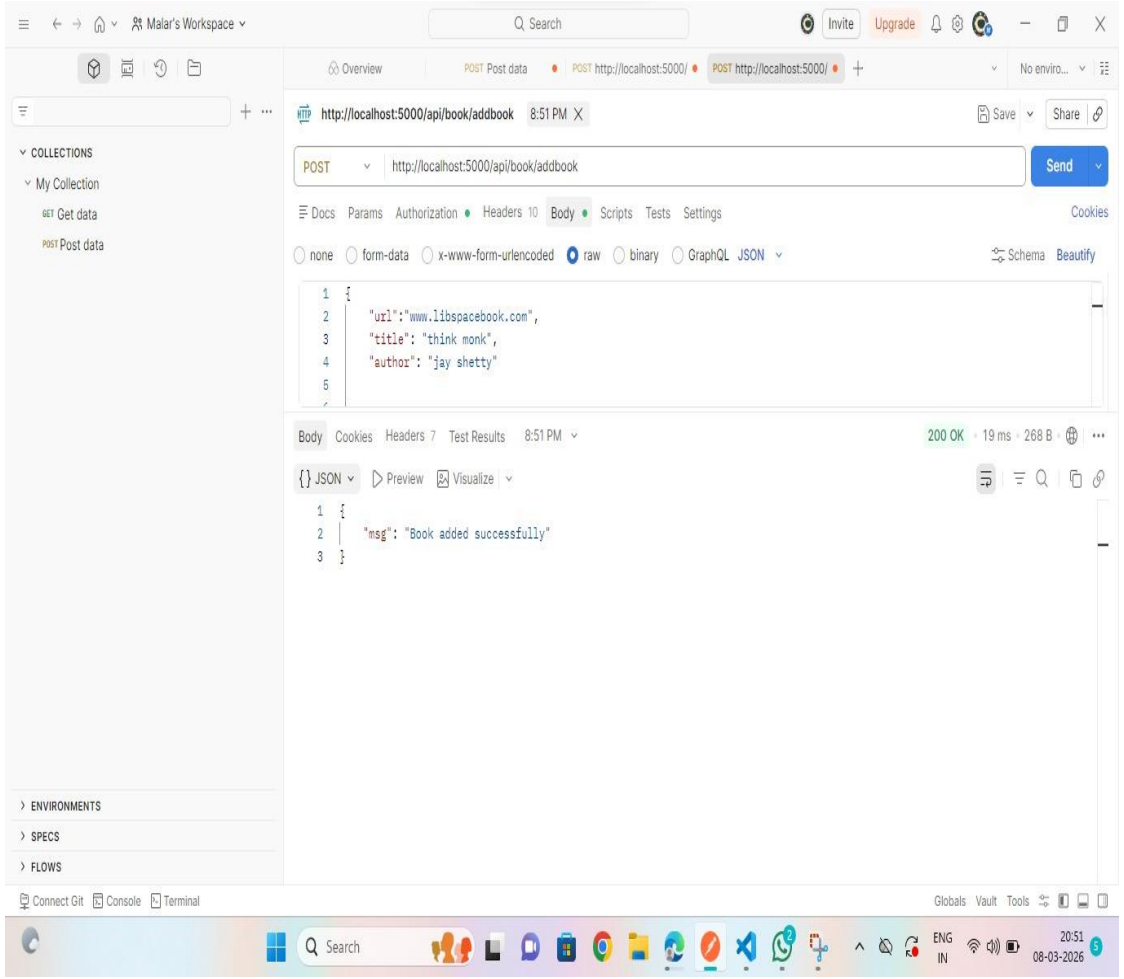
User registration



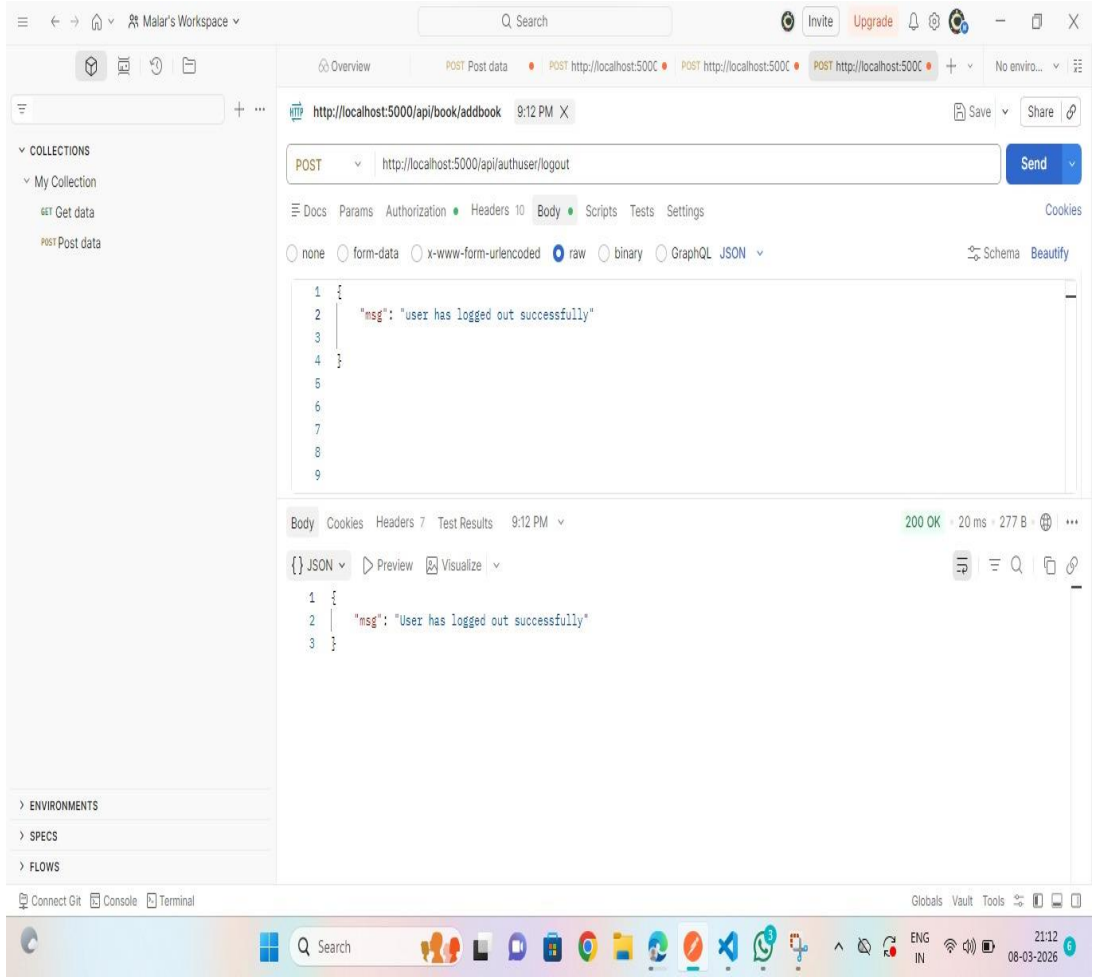
User Login



Adding book



User Logout



Future Enhancements:

- Mobile application support.
- Barcode scanning system.
- Cloud storage integration.
- Al-based book recommendation.

Conclusion:

- LibSpace modernizes library management.
- Improves productivity and accuracy.
- User-friendly and efficient system.
- Suitable for schools and colleges.